Lab 7 - Fuzzing for Input Validation Bugs

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1. Original application

The original script returned a cart total of 270, a discount of 20%, and the South zone for the sample inputs.

```
> python shopping_cart.py
Input: pen:10x2,book:250x1 | OFF20 | 560085
==== Running cart_total ====
Enter cart as name:pricexqty CSV (e.g. pen:10x2,book:250x1): Cart Total: 270

==== Running promo_percent ====
Enter a promo code (e.g. OFF20): Discount Percent: 20

==== Running delivery_zone ====
Enter a 6 digit pincode (e.g. 560085): Delivery Zone: South
```

2. Tests before the fixes

Hypothesis generated text, missing values, integers and bytes. The three no-crash checks each failed on the original script.

```
> python -m pytest -q --tb=no -k no_crash test_processor.py
FFF                                                                      [100%]
=============================================== short test summary info ===============================================
FAILED test_processor.py::test_cart_total_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-excep...
FAILED test_processor.py::test_promo_percent_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-ex...
FAILED test_processor.py::test_delivery_zone_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-ex...
3 failed, 21 deselected in 0.90s
```

Cart. A blank cart raised ValueError during splitting. It now totals 0. Each item must have a name, a non-negative integer price and a positive integer quantity.

Promo. Blank input raised IndexError. The parser now requires OFF followed by a percentage from 0 to 100 and accepts either letter case.

Delivery. Blank input raised IndexError. The parser now checks for six ASCII digits and a supported first digit from 1 to 8.

Source: SETAPESU26/L6_L7_23_shopping_cart, Lab 7.

3. Fixed application

The same sample inputs still return 270, 20% and South. Invalid inputs are rejected with a message, and the program continues to the next prompt.

```
> python shopping_cart.py
Input: pen:10x2,book:250x1 | OFF20 | 560085
==== Running cart_total ====
Enter cart as name:pricexqty CSV (e.g. pen:10x2,book:250x1): Cart Total: 270

==== Running promo_percent ====
Enter a promo code (e.g. OFF20): Discount Percent: 20

==== Running delivery_zone ====
Enter a 6 digit pincode (e.g. 560085): Delivery Zone: South

> python shopping_cart.py
Input: pen:10x0 | OFF101 | 56008A
==== Running cart_total ====
Enter cart as name:pricexqty CSV (e.g. pen:10x2,book:250x1): Invalid cart. Check the item names, prices and quantities.

==== Running promo_percent ====
Enter a promo code (e.g. OFF20): Invalid promo code. Enter OFF followed by a number from 0 to 100.

==== Running delivery_zone ====
Enter a 6 digit pincode (e.g. 560085): Invalid pincode. Enter six digits starting with 1 to 8.
```

4. Tests after the fixes

The full suite passed all 24 tests. It includes six Hypothesis properties with up to 150 examples each and 18 cases that check invalid inputs.

```
> python -m pytest -q test_processor.py
........................                                                 [100%]
24 passed in 0.66s
```

Run these commands from the lab7 folder:

```
python -m pip install hypothesis pytest
python shopping_cart.py
python -m pytest -q test_processor.py
```